

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

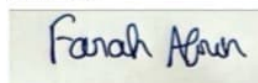
Concept Mapping
Annexure A

Criteria	Concept Identification (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I M. Farah Afrin / 1925114343 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

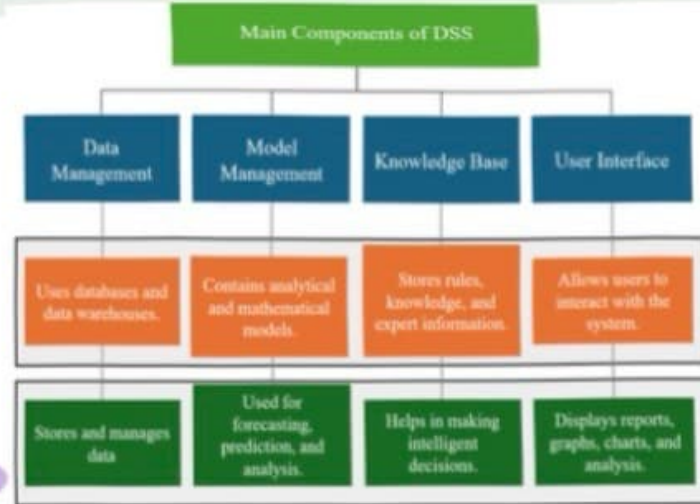
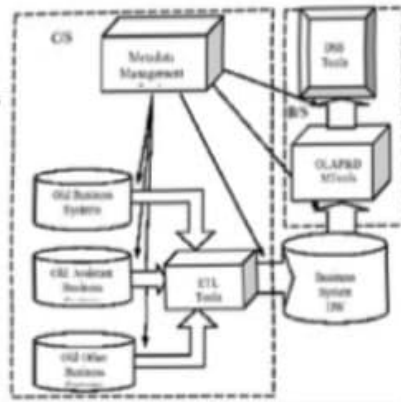
Decision Support System

A Decision Support System (DSS) is a computer-based system that helps managers and decision-makers make better decisions by analysing data and providing useful information.

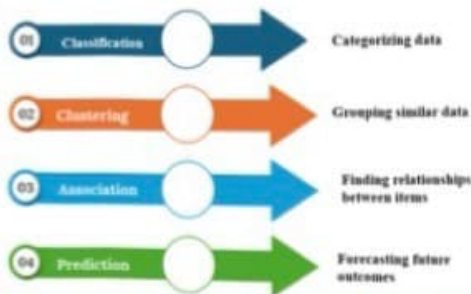
A DSS collects data from different sources, analyzes it, and presents the results in a meaningful way to support semi-structured and unstructured decision-making.

Example: A supermarket analyzes customer purchase data to decide product stock, pricing, and marketing strategies.

Benefits: Faster decisions • Better analysis • Forecasting • Reduced risk



Role of Data Mining in decision support system



Role of Datawarehousing in decision support system



OLAP - OLAP (Online Analytical Processing) is a technology used to quickly analyze large amounts of data from different perspectives for decision making.

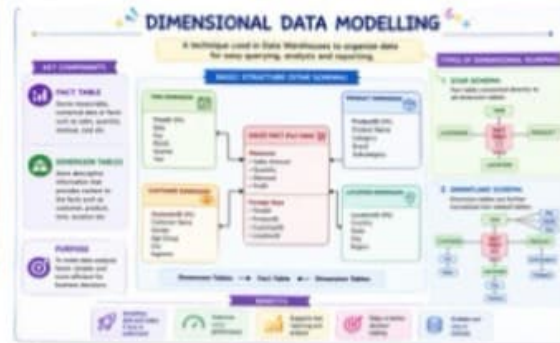
Roll-up	Summarizes data (Year → Quarter → Month)
Dive - Drill	Shows detailed data (Year → Month → Day)
Slice	Selects one dimension (Sales in 2020)
Dice	Selects multiple dimensions (Sales of laptops in Chennai during 2020)
Pivot	Rotates the data view for a different perspective

Data Cube:
A Data Cube is a multidimensional representation of data that allows users to analyze information across different dimensions.
Example: Sales Data Cube

Dimensions:
Time • Product • Location

Measures:
Sales Amount • Quantity

Sold • Profit



Operational System	Decision Support System (DSS)
➤ Used for day-to-day operations	➤ Used for analysis and decision-making
➤ Handles current, detailed data	➤ Uses historical and summarized data
➤ Supports routine and repetitive tasks	➤ Supports complex and strategic decisions
➤ Data is frequently updated	➤ Data is analyzed for trends and patterns
➤ Example: ATM, billing, order processing	➤ Example: Sales forecasting and business analytics



Data Warehouse → OLAP Application → Multidimensional Cube → Analysis & Reports

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CSAI603 Datawarehousing and Data Mining
Assessment Tool - 2